



# Cellular Adaptation



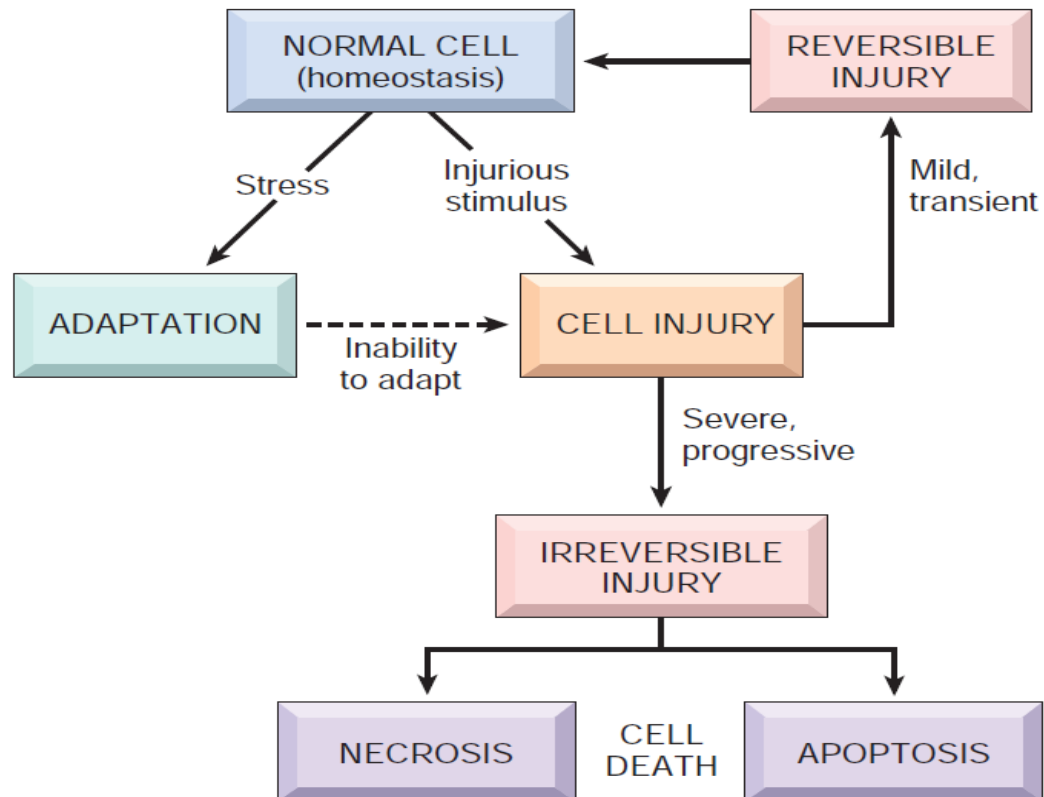
Bethel F.

# Introduction

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- ▶ Cells are normally maintaining a healthy steady state called *homeostasis*.
- ▶ ***Adaptations*** are reversible functional and structural responses to changes in physiologic states (e.g., pregnancy) and some pathologic stimuli.





**Figure 2.1** Stages of the cellular response to stress and injurious stimuli.

# Cellular Adaptation

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- ▶ are **reversible changes** in the number, size, phenotype, metabolic activity, or functions of cells in response to changes in their environment.
- ▶ Cells adapt to changes in the internal environment by undergoing changes in size, number, and type.
- ▶ Cells adapt to increased work demands by changing in
  - ▶ Size (atrophy and hypertrophy)
  - ▶ Number (hyperplasia)
  - ▶ Form (metaplasia)



# Hypertrophy

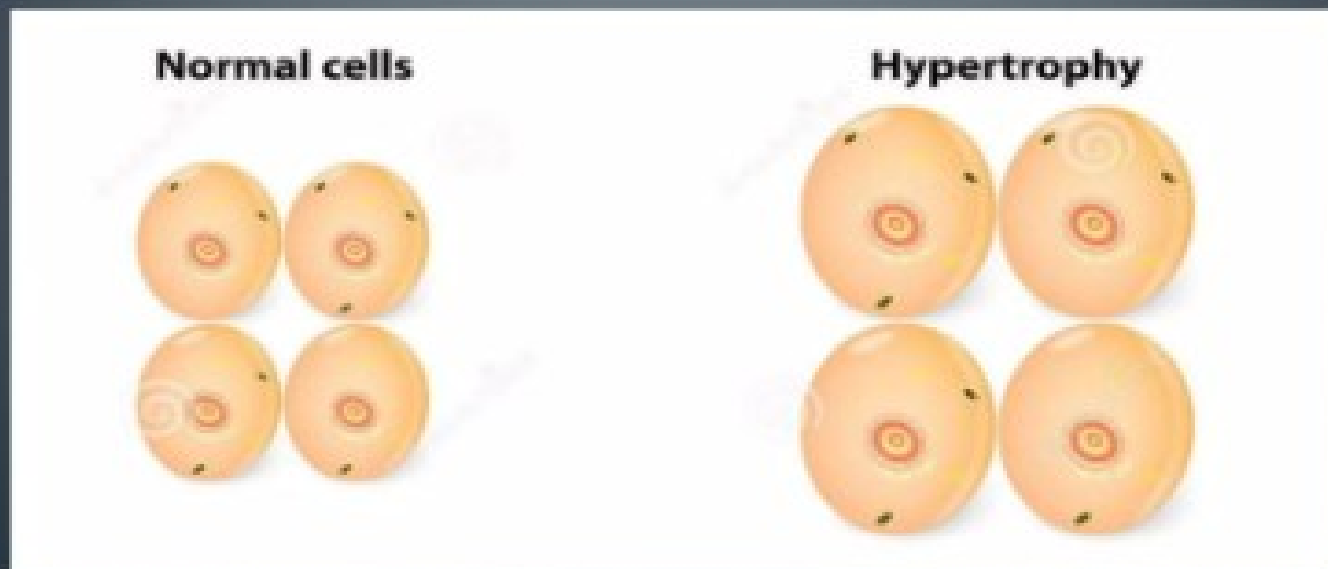
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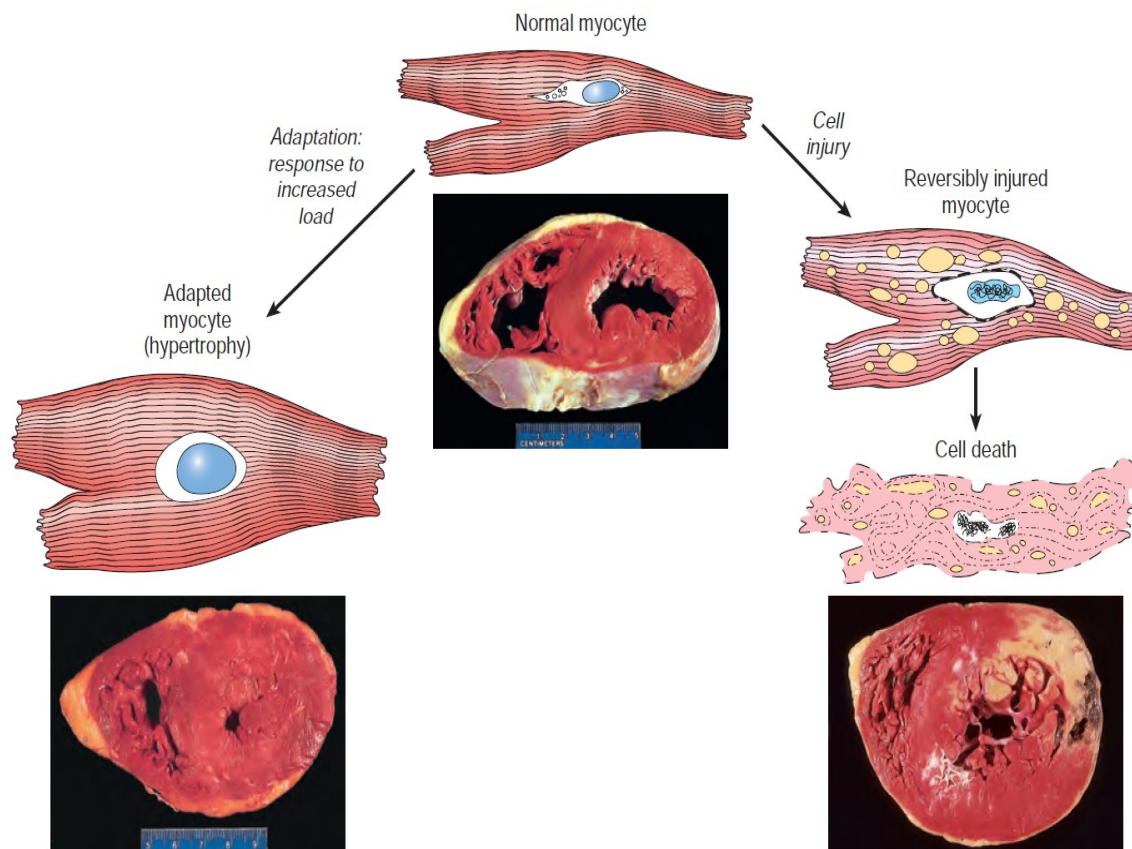
- ▶ is an **increase in the size of cells resulting in increase in the size of the organ.**
- ▶ in pure hypertrophy there are no new cells, just bigger cells, enlarged by an increased amount of structural proteins and organelles.
- ▶ can be physiologic or pathologic and is caused either by increased functional demand or by specific hormonal stimulation.



# HYPERTROPHY

- Hypertrophy is an increase in the size of parenchymal cells resulting in enlargement of the organ or tissue, without any change in the number of cells.





# Hyperplasia

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- ▶ is characterized by an increase in cell number.
- ▶ is an adaptive response in cells capable of replication
- ▶ **Hypertrophy and hyperplasia** can also occur together, and obviously both result in an enlarged (hypertrophic) organ.



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- ▶ There are two types of physiologic hyperplasia
    - ▶ hormonal hyperplasia, exemplified by the proliferation of the **glandular epithelium of the female breast at puberty and during pregnancy**;
    - ▶ compensatory hyperplasia, that is, hyperplasia that occurs when a portion of the tissue is removed or diseased.
      - ▶ **E.g liver**



# Atrophy

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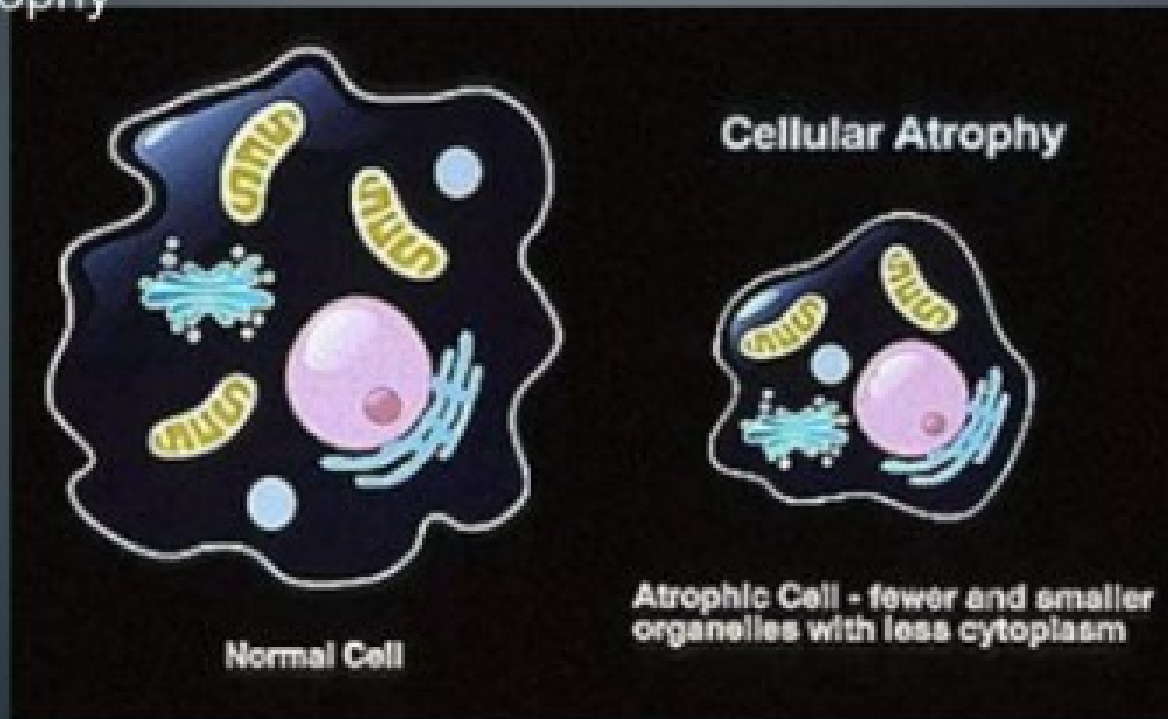
- ▶ Is shrinkage in the size of the cell by the loss of cell substance
- ▶ When a sufficient number of cells is involved, the entire tissue or organ diminishes in size, becoming atrophic
- ▶ **atrophic cells may have diminished function, they are not dead.**
- ▶ **Causes:-malnutrition, disuse, neuronal damage.....**



# Cellular Adaptation

## ATROPHY

Reduction of the number and size of parenchymal cells of an organ or its parts which was once normal is called atrophy



# Metaplasia

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- ▶ is a reversible change in which one adult cell type (epithelial or mesenchymal) is replaced by another adult cell type.
- ▶ In this type of cellular adaptation, cells sensitive to a particular stress are replaced by other cell types better able to withstand the adverse environment.



| Tissue          | Normal                               | Metaplasia                                   | Stimulus                 |
|-----------------|--------------------------------------|----------------------------------------------|--------------------------|
| Airways         | Pseudostratified columnar epithelium | Squamous epithelium                          | Cigarette smoke          |
| Urinary bladder | Transitional epithelium              | Squamous epithelium                          | Bladder stone            |
| Esophagus       | Squamous epithelium                  | Columnar epithelium ( Barrett's Esophagu ... | Gastro-esophageal reflux |
| Cervix          | Glandular epithelium                 | Squamous epithelium                          | Low pH of vagina         |

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# CELL INJURY



Almost all forms of diseases start with molecular or structural alterations in cells

Injury to cells and extracellular matrix ultimately leads to tissue and organ injury

We will discuss about  
Cause , mechanisms , morphology and  
biochemical alteration of cell injury



# Cellular Responses To Injury

## 1. Cellular Adaptations

### Atrophy, Hypertrophy, Hyperplasia & Metaplasia

## 2. Acute cell injury

Reversible cell injury -> Degenerations

Irreversible cell injury -> Cell death – Necrosis  
AND / OR apoptosis

## 3. Subcellular alterations & cell inclusions

## 4. Intracellular accumulations

## 5. Pathologic calcifications



Normal cell : is in a steady state of homeostasis according to the physiologic demands.

1. Adaptations : *reversible functional and structural responses to more severe physiologic stresses and some pathologic stimuli.*

→ new altered steady state that favors survival and functionality.



- ▶ If limits of adaptive responses are exceeded , cells are exposed to injurious agents or stress → sequence of events follow → **cell injury**.

2. Cell injury → **reversible** up to certain point

- ▶ Persistent stimulus , severe stimulus from out set → **irreversible cell injury** → cell death

- ▶ Two principal pathways of cell death = > **necrosis** and **apoptosis**

3. Metabolic derangements and sublethal, chronic injury → **intracellular accumulations**
4. Calcium often deposited at sites of cell death → **pathologic calcification**
5. Subcellular alterations & cell inclusions



# Overview of cell injury and cell death

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\* Cell injury results when :

- ▶ Severe stress **beyond adaptation**
- ▶ **Inherently** damaging agents
- ▶ **Intrinsic abnormalities**

- May progress through reversible stage and end in cell death.



# Reversible cell injury

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- ▶ Early stage, mild form
  - ▶ Functional and morphologic features are **reversible** if stimulus removed
  - ▶ Characterized by :
    - ▶ A. **reduced** oxidative phosphorylation ( ↓ **ATP** ) ,
    - ▶ B. **cellular swelling** ,
    - ▶ C. **mitochondrial and cytoskeletal alterations**
- 



# Reversible and Irreversible Cell Injury

- ▶ **Transition** from reversible to irreversible injury is **unclear**
- ▶ Inter-dependence of biochemical systems in the cell makes it difficult to determine the exact sequence of changes
- ▶ Two features are characteristic of irreversibility
  - ▶ **Irreversible mitochondrial dysfunction**
  - ▶ **Profound membrane dysfunction**

## Cell death

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- ▶ Continuous damage = > irreversible injury  
=> death
- ▶ Necrosis and apoptosis → different morphology , mechanism , role in physiology and diseases.



# Causes Of Cell Injury

- ▶ **HYPOXIA** - deficiency of oxygen  
*ischemia , inadequate oxygenation of the blood , loss of oxygen carrying capacity of the blood*
  - ▶ **Physical Agents**  
Trauma , radiation , extremes of temperature, electric shock, sudden change in atmospheric pressure
  - **Chemicals:** Drugs (e.g. paracetamol), poisons (e.g. paraquat, cyanide)
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# Causes Of Cell Injury

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- ▶ Infectious agents
- ▶ Immunologic agents
- ▶ Genetic Derangements
- ▶ Nutritional imbalances

Protein-energy Malnutrition, Vitamin deficiencies & Nutritional excesses



# Biochemical Site Of Action Of Injurious Agents

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- ▶ Whatever the injurious agent, cell injury results when there is **dysfunction of essential cellular systems**:
- ▶ Maintenance of cell membranes
- ▶ Aerobic respiration
- ▶ Enzymatic & structural proteins
- ▶ Genetic apparatus



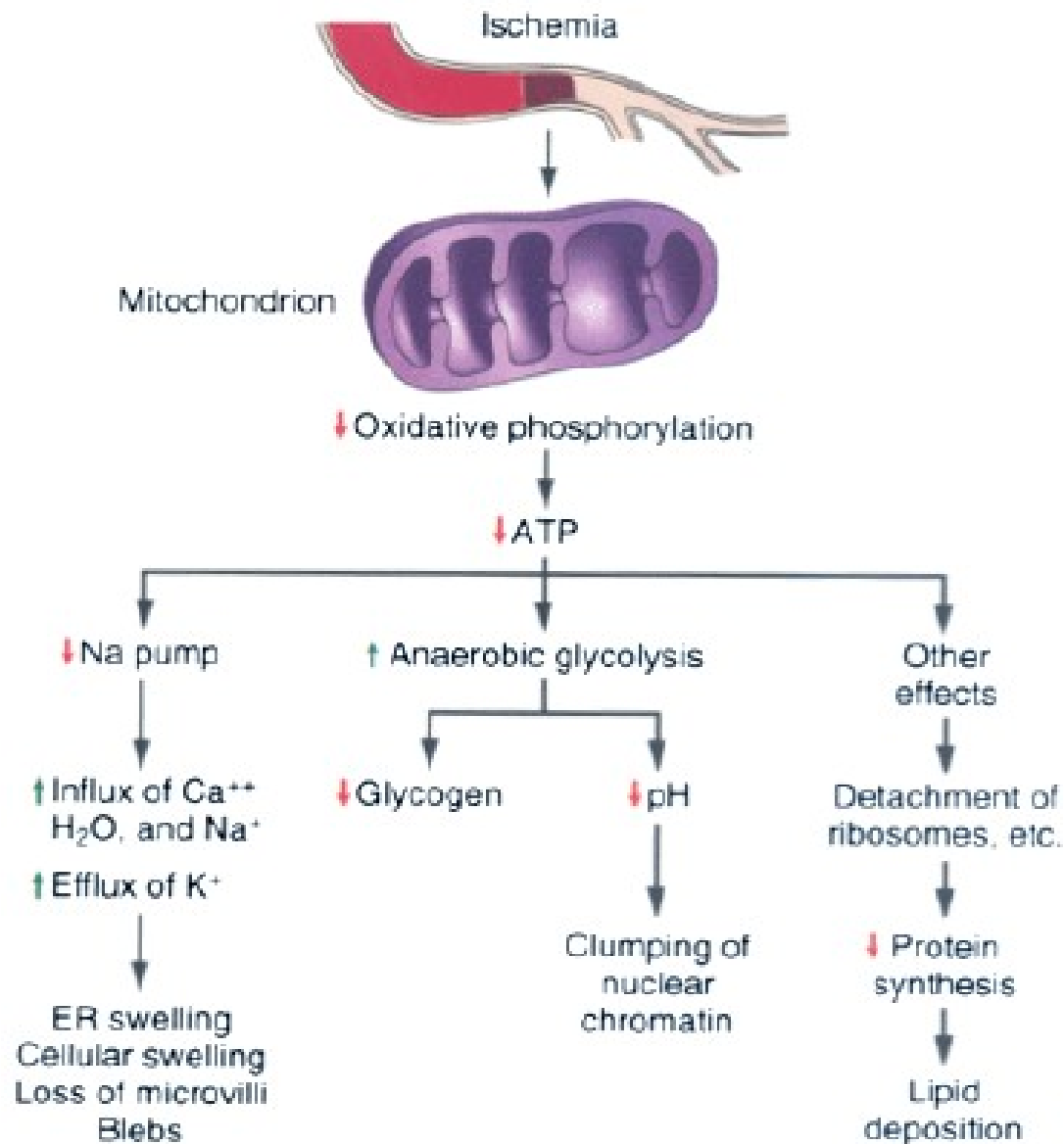
# Mediators / Mechanisms Of Cell Injury

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- ▶ Oxygen & oxygen- derived free radicals
- ▶ Intra cellular calcium
- ▶ ATP depletion
- ▶ Defects in membrane permeability



2. Mitochondrial dysfunction caused by reduced supply of oxygen, nutrients, mitochondrial damage, toxins



## 2. Accumulation of Oxygen & oxygen-derived free radicals (**oxidative stress**)

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- ▶ *Chemical and radiation injury, ischemia - reperfusion injury , microbial killing*
- ▶ **Free radicals** : single unpaired electron on outer orbit
- ▶ **Reactive oxygen species** : oxygen derived free radicals
- ▶ Production and removal is balanced , with no damage.
- ▶ Increased production and decreased clearance => **oxidative stress**

# Generation of free radicals

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- ▶ Reduction - oxidation reaction of normal metabolic process
  - ▶ Absorption of radiant energy
  - ▶ Activated leuckocytes during inflammation
  - ▶ Enzymatic metabolism - drugs and chemicals
  - ▶ Transition metals
  - ▶ Nitric oxide
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# Removal of free radicals

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- ▶ Spontaneous decay
- ▶ Antioxidants
- ▶ Free radical scavenging systems of enzymes

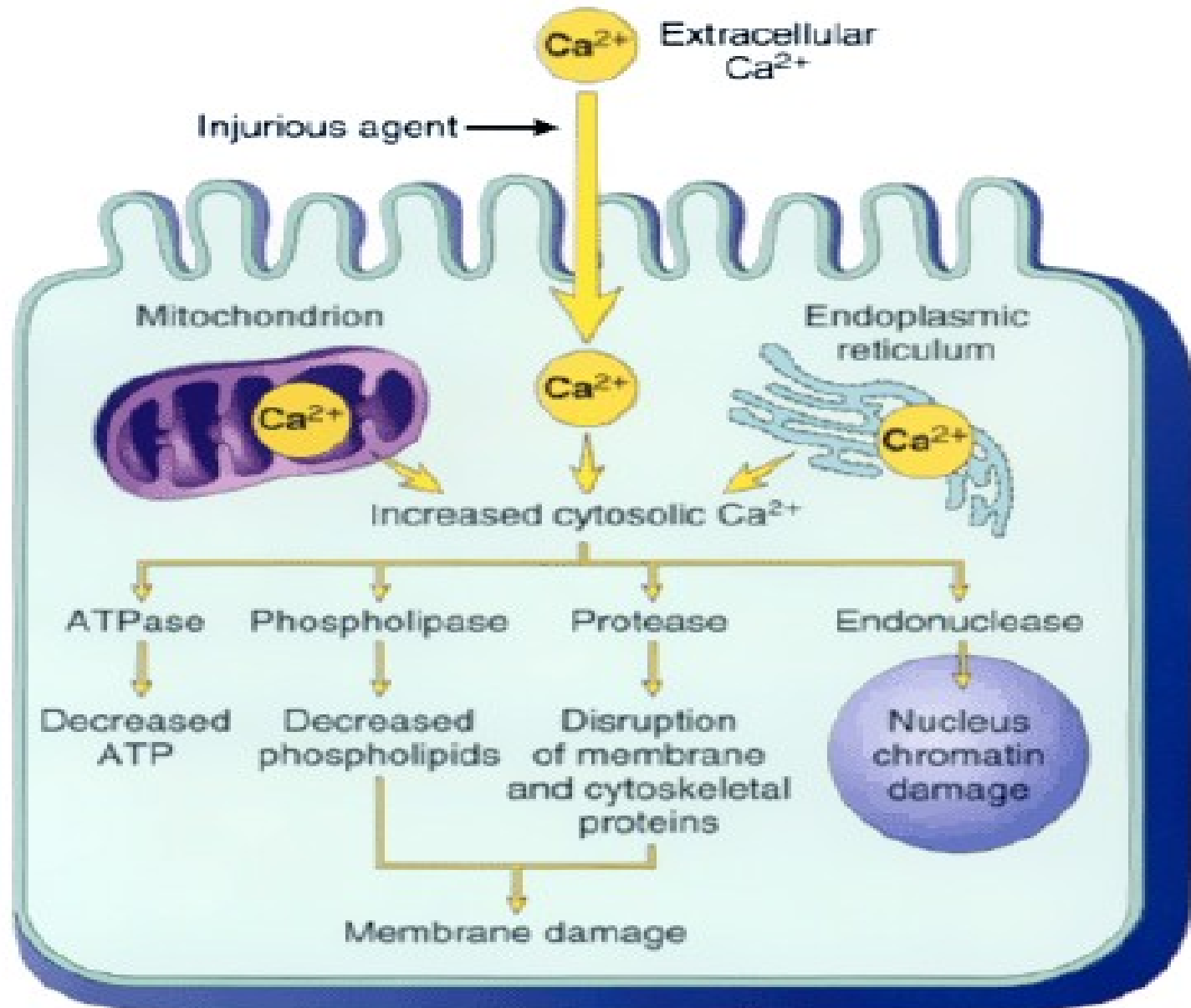


# Pathological effects of free radicals.

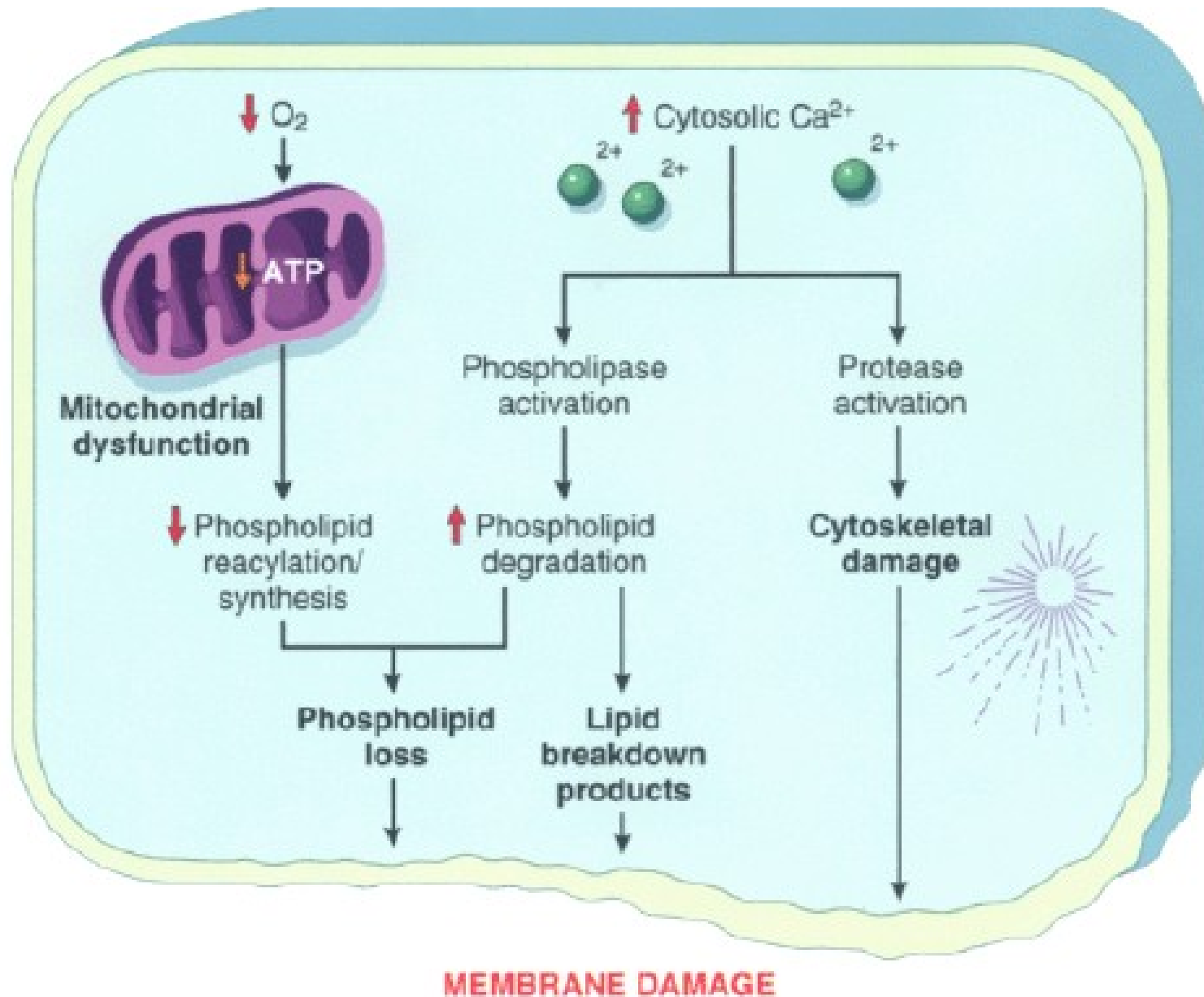
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- ▶ **Lipid peroxidation in membranes** => auto catalytic reaction with extensive membrane damage
  - ▶ **Oxidative modification of proteins** => active site of enzyme damage , disruption of structural proteins
  - ▶ **DNA damage** => breaks in DNA , cross linking of strands
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### 3. Increased intracellular calcium



## 4. Defects in membrane permeability



## Out comes of membrane damage

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- ▶ *Mitochondrial membrane damage* => decreased ATP
- ▶ *Plasma membrane damage* => loss of cellular contents and osmotic balance.
- ▶ *Lysosomal membrane damage* => leakage of enzymes



# Morphological Changes

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- ▶ Cell injury can be seen morphologically (by light microscopy) as
    - ▶ **Hydropic change** (Reversible cell injury)
    - ▶ **Fatty change** (Reversible cell injury)and
    - ▶ **Necrosis** (Irreversible cell injury)
    - ▶ **Apoptosis** (Irreversible cell injury)
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# APOPTOSIS

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It is a form of cell death induced by tightly regulated suicide program in which the cell activates enzymes that degrade the cells own nuclear DNA and nuclear and cytoplasmic proteins.



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Apoptotic cells break up into fragments = >  
**apoptotic bodies** (contain portion of nucleus  
and cytoplasm)

**Structural alteration** of membranes  
attracting phagocytes ( **cell content does  
not leak out**)

Programmed cell death

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# Causes of apoptosis

## Physiologic

**Embryogenesis** : implantation , organogenesis, developmental involution

**Involution of hormone dependent structures** up on withdrawal : regression of breast tissue after weaning

**Cell loss** in proliferating cell population : to **maintain homeostasis**=> in lymphnode

**Elimination of self reacting** lymphocytes

Death of cells **after fulfilling** their function



# Pathologic

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**DNA damage** : beyond repair => radiation, hypoxia , chemicals ..... risk of malignancy

**Viral infections** ( HIV, HBV, adenovirus)



# Occurrence Of Apoptosis

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One of the **morphologic patterns** of cell death

Plays a role in the development of an embryo , metamorphosis & regulation of tissue size throughout life



# Occurrence Of Apoptosis

## Morphogenetic Apoptosis

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Inter digital cell death - syndactylyl

Cell death leading to the removal of redundant epithelium following fusion of the palatine processes - cleft palate

Cell death in the dorsal part of the neural tube - spinal bifida

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Webbed foot a classic example of failure of apoptosis.

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# Occurrence Of Apoptosis

## Phylogenetic Apoptosis

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Removal of vestigial structures from the embryo

⇒ **In the fetus**: the Branchial cleft ,  
notochord & the thyroglossal duct,

⇒ **In infancy** \_ the ductus arteriosus &  
umbilical vessels,

⇒ **Later life** \_ lymphoid tissue, menopause  
& in old age

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# Apoptosis in physiologic, adaptive & pathologic events : **Summary**

hormone dependent involution  
cell death in proliferating cell population  
cell death in tumor cells  
death in immune cells  
pathologic atrophy after duct obstruction  
Cell injury in viral diseases



# Morphology

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- ▶ Apoptosis

  - => Death of **single cells** in tissue

  - => **No inflammatory** reaction

- ▶ Necrosis

  - => death of **large amount of cells** in tissue at the same time

  - => **inflammatory reaction**

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# Morphology

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1. Cell shrinkage
2. Chromatin condensation
3. Cytoplasmic blebs & apoptotic bodies
4. Phagocytosis of apoptotic cells or bodies by macrophages

In contrast to necrosis, apoptosis does not elicit inflammation



# Biochemical Features

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Protein cleavage \_ protein hydrolysis  
involving the activation of proteases

Protein Cross-linking\_ by activation of  
transglutaminase

DNA Breakdown \_by activation of  
endonucleases

Phagocytic Recognition

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# Deranged Apoptosis

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Increased Apoptosis is seen in

- ▶ Viral Infection
- ▶ Graft rejection
- ▶ Neurodegenerative disorders

Decreased Apoptosis is the seen in

- ▶ Neoplasia
- ▶ Autoimmune disorders



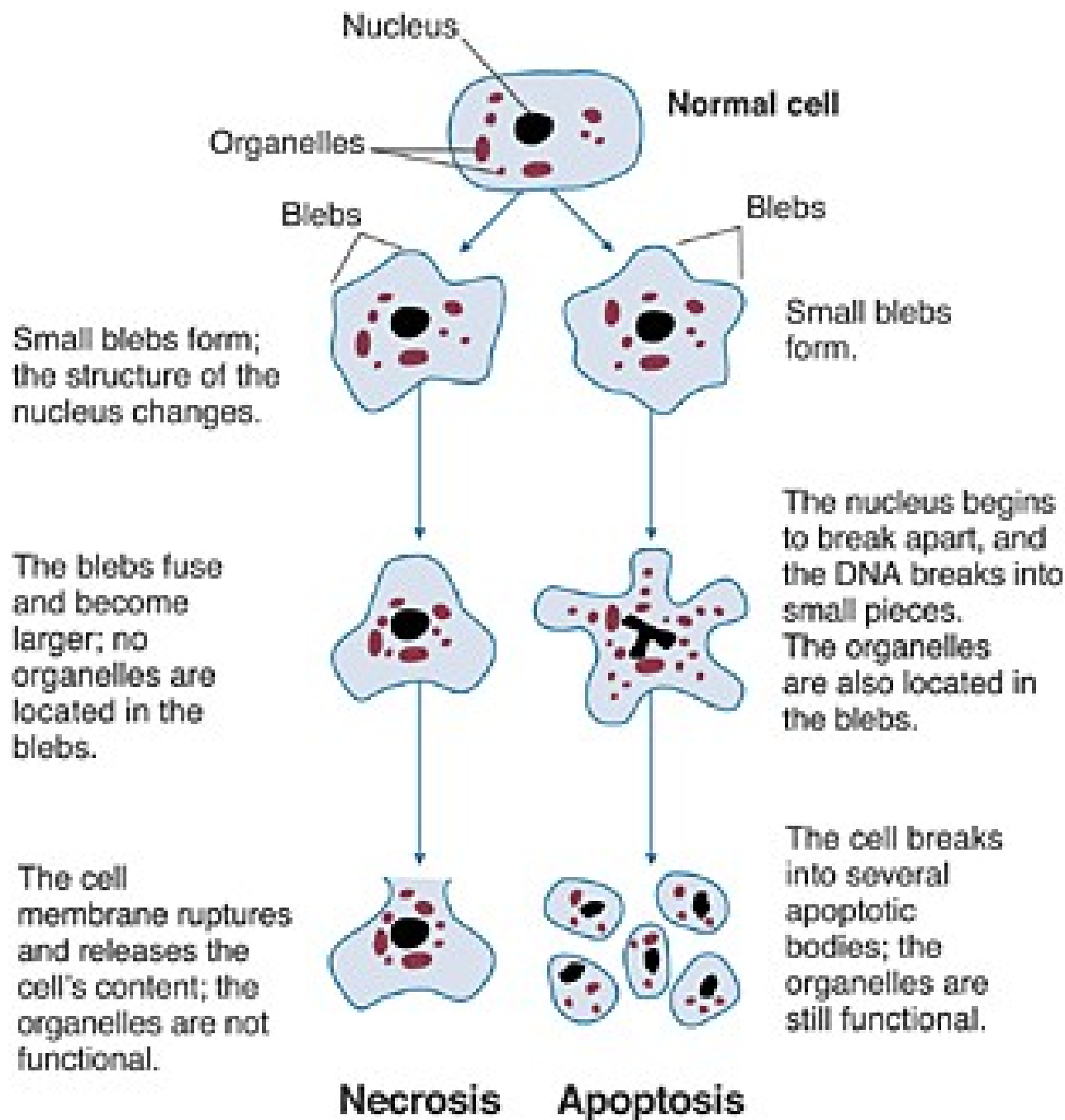
## Differing Features

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Apoptosis => Gene Activation =>  
Endonucleases

Tissue Reaction: Necrosis-inflammation  
: Apoptosis => no inflammation,  
Phagocytosis of apoptotic bodies





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# NECROSIS

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- ▶ Necrosis refers to a spectrum of morphologic changes that follow cell death in living tissue
- ▶ Necrosis is the gross & histologic correlate of cell death occurring in the setting of irreversible exogenous injury
- ▶ Cells immediately placed in fixative are dead but not necrotic



# Necrosis

Cell necrosis should be distinguished from the death of the individual

The morphologic appearance of necrosis is the result of two essentially concurrent processes (1) **enzymic digestion of the lethally injured cell** and (2) denaturation of proteins.

Loss of membrane integrity in necrotic cells => leakage of chemicals => **inflammation.**

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Enzymes digesting the cells are derived either from the lysosomes of the dead cells → **Autolysis**

or

from the lysosomes of immigrant leukocytes → **Heterolysis**



# Necrosis

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These processes ( digestion of the cell and host response ) require **hours to develop**  
→ gross and microscopic changes **not evident immediately.**



# Morphology of necrotic cells (light M)

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**Increased eosinophilia** - loss of cytoplasmic RNA and denaturation of protein

**Homogenous glassy appearance** : loss of glycogen

**Vacuolated cytoplasm** with moth-eaten appearance



# Nuclear changes

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**Karyolysis** : fading of chromatin  
basophilia => DNA breakdown by  
endonuclease

**Pyknosis** : nuclear shrinkage and increased  
basophilia => chromatin condense in to  
solid shrunken mass.

**Karyorrhexis** : fragmentation of pyknotic  
nuclei.

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# Patterns / Types Of tissue Necrosis

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Necrosis of large number of cells =>  
tissue / organ necrosis

➤ Morphologically distinctive patterns =>  
cause can be identified

➤ Determined by

1. Variation in cell composition
2. Speed of necrosis
3. Type of injury



# Necrosis

## 1. Coagulative necrosis-

The most common

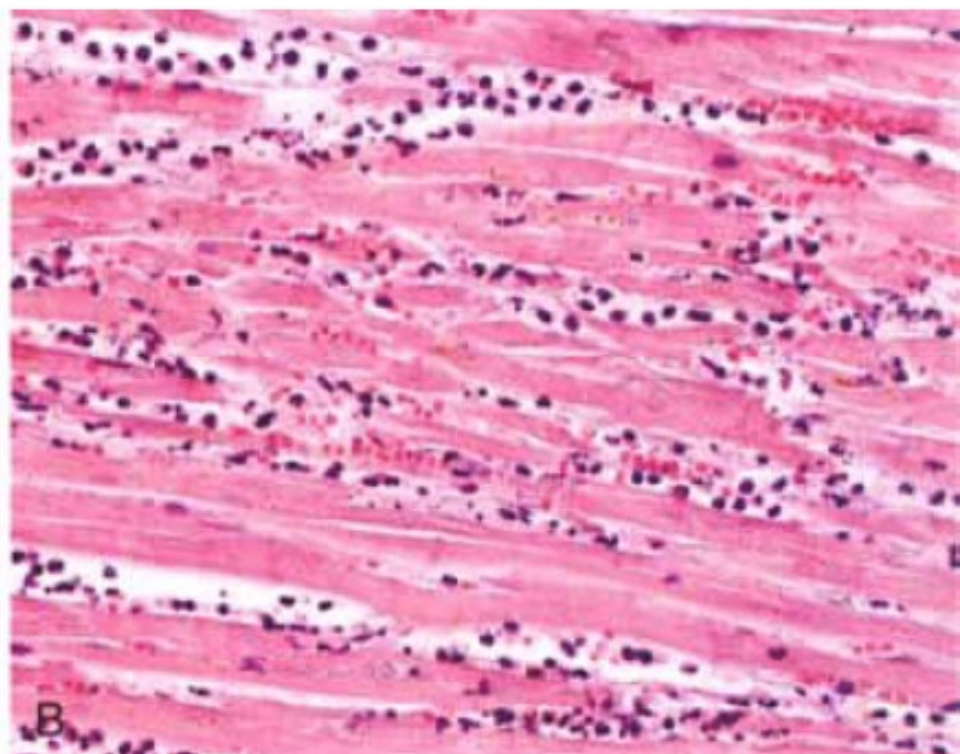
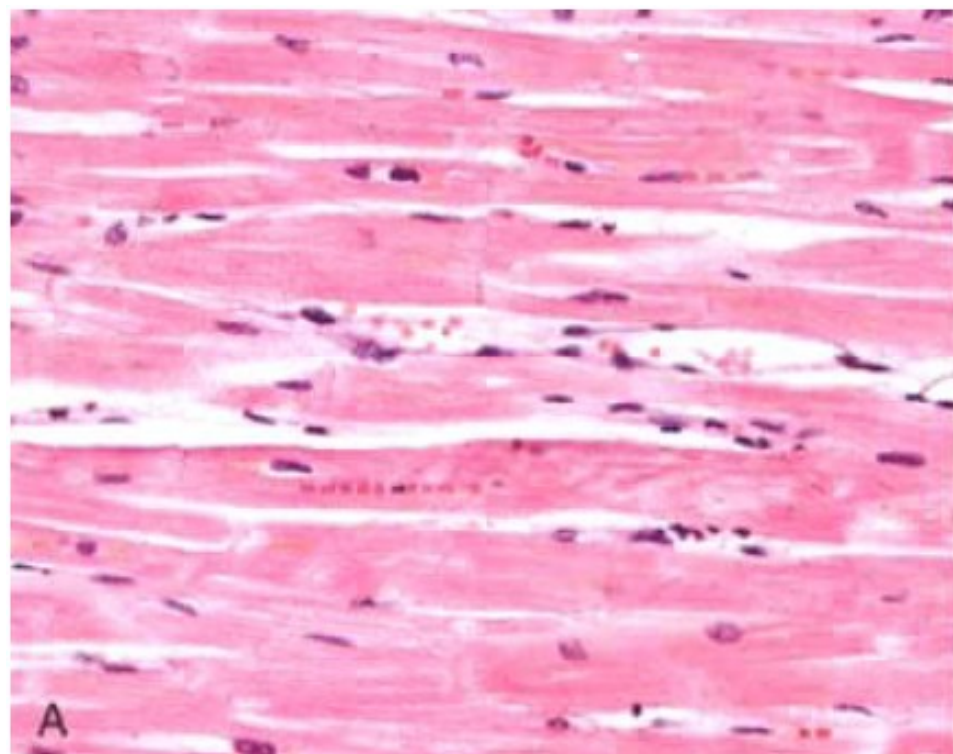
Preservation of architecture of dead tissue for some days

Denaturation of structural and enzymatic proteins

Ischemia of all solid organs except brain  
toxic chemicals, burns, viral infections

Infarct - localized area of coagulative necrosis





## 2. Liquefactive necrosis

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Digestion of dead tissue = > liquid viscous mass : usually has creamy yellow appearance

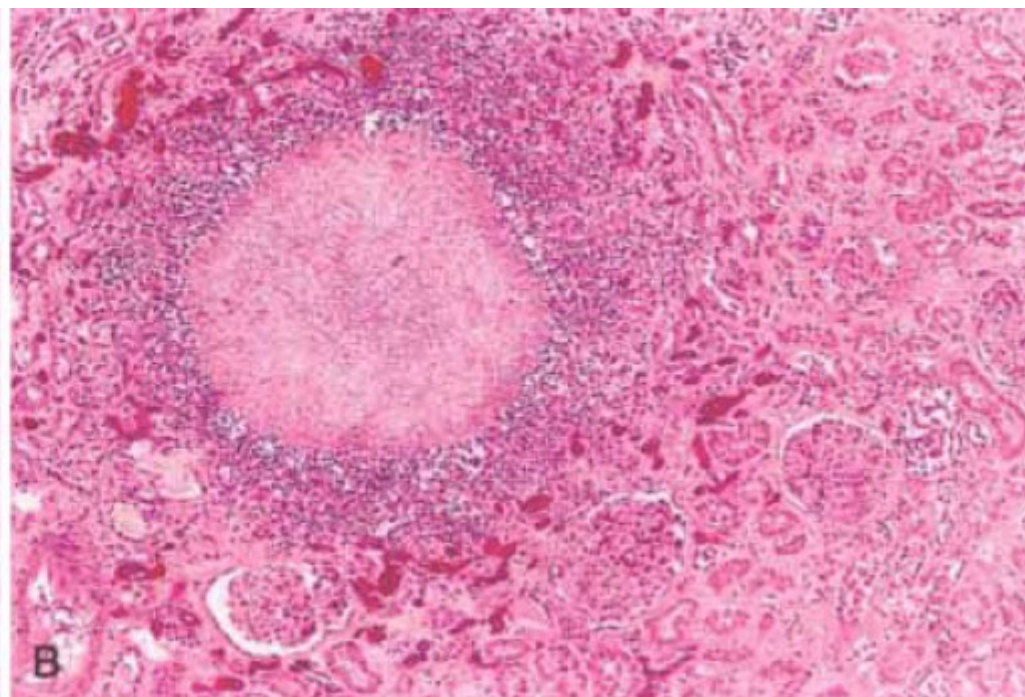
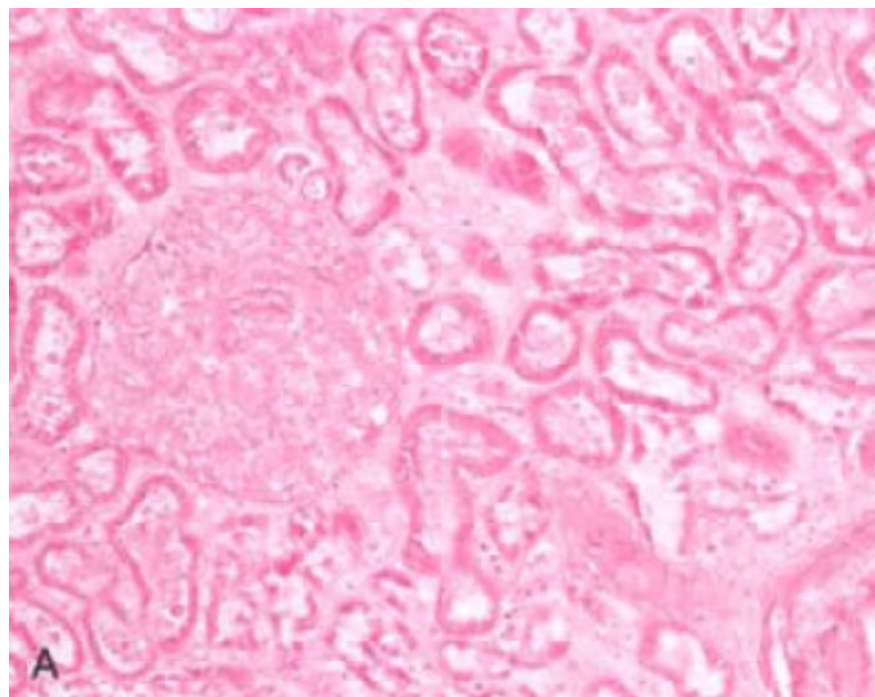
= > Pus

Mechanism - Autolysis or Heterolysis  
Bacterial or Fungal infections (Pus formation)  
)

Brain following ischemia

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### 3 Fat necrosis-

Focal area of fat destruction

**Enzymatic or non enzymatic**

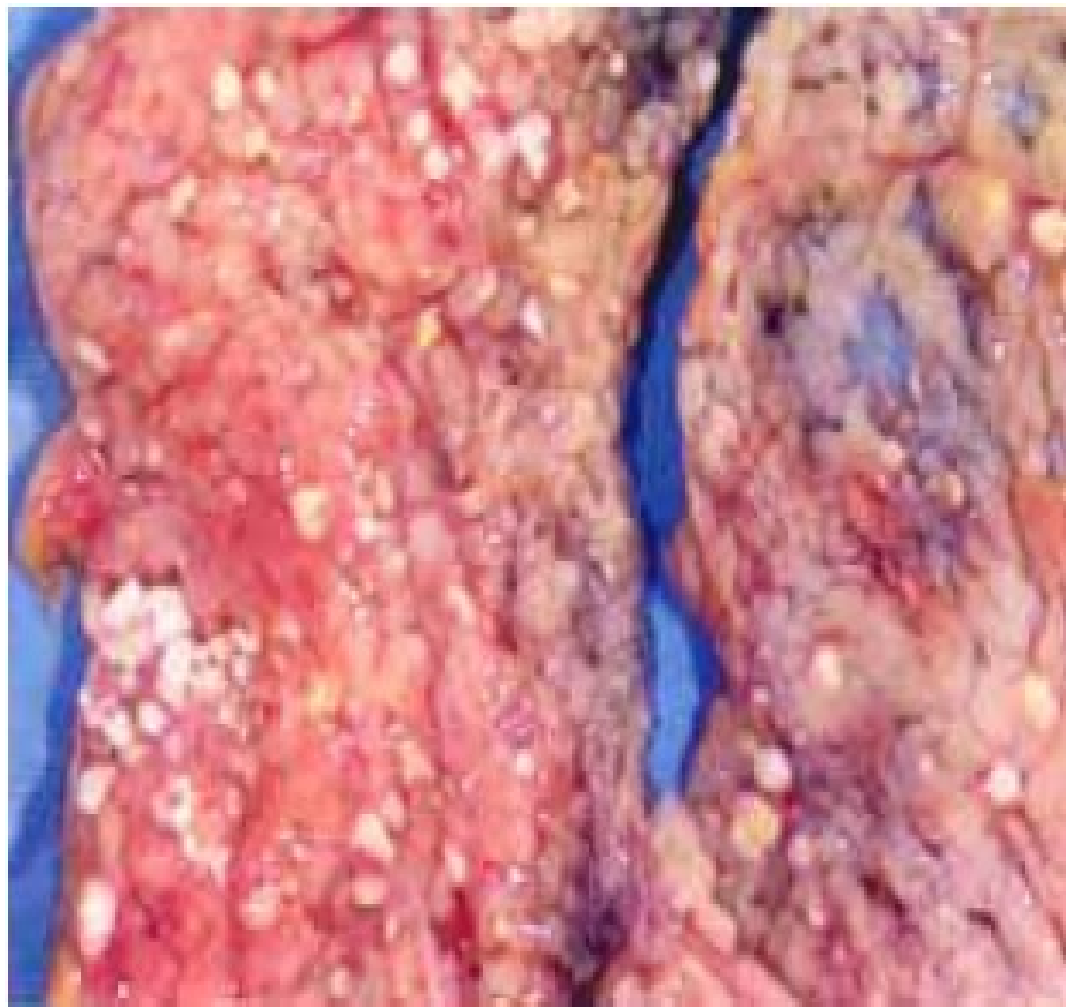
**Enzymatic : Acute pancreatitis** : leakage of enzymes from pancreas which lyses cell membrane of fat cells in peritoneum

**Lipase** : split triglycerides to FA which combine with calcium => **chalky white** areas (**fat saponification**) visible grossly.

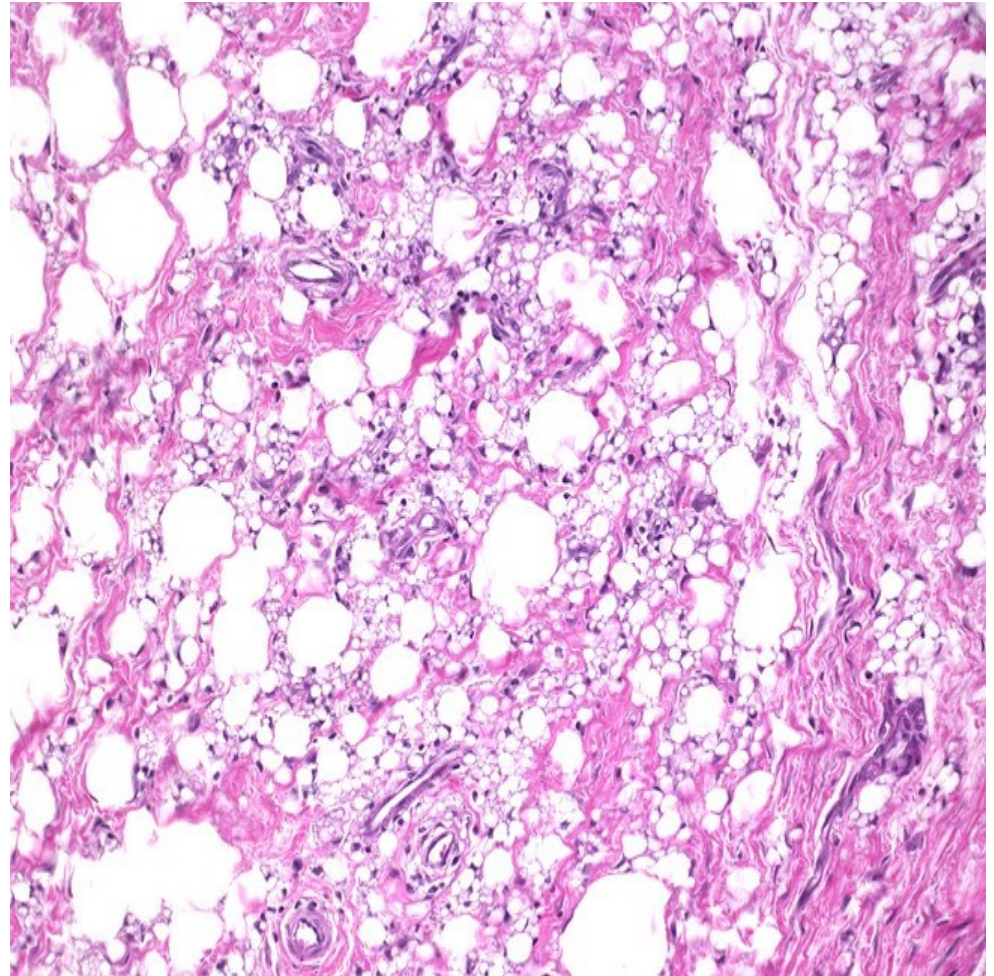
**Non enzymatic fat necrosis** :

Breast ,subcutaneous fat due to **trauma**





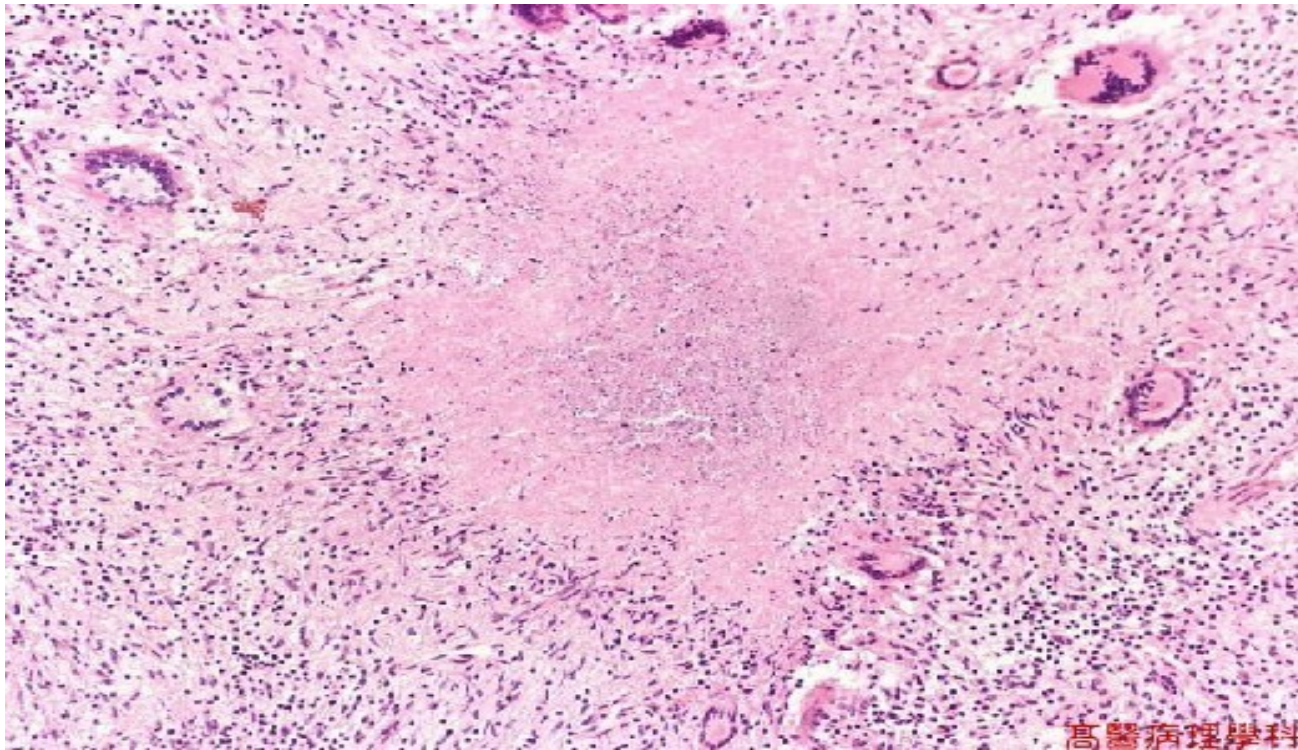
On histologic examination:  
The necrosis takes the  
form of foci of shadowy  
outlines of necrotic fat  
cells, with basophilic  
calcium deposits,  
surrounded by an  
inflammatory reaction



## 4. Caseous necrosis

occur in foci of tuberculosis infection

**Friable** , **white** area of necrosis looks like cheese.



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## 5. Gangrenous necrosis

Clinical term

- ▶ Limb, usually lower , coagulative necrosis due to ischemia => **dry gangrene**
- ▶ Superimposed bacterial infection => liquifactive necrosis = > **wet gangrene**



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applied to any black, foul-smelling area that is in continuity with living tissue.

It is necrosis of tissue with superadded putrefaction e.g. Dry, Wet, and Gas gangrene

Bowel obstruction with necrosis



# Necrosis

## 5 Fibrinoid necrosis:

**Immune** reactions involving **blood vessels**

Deposition of antigen - antibody complexes in arterial wall

Deposited complexes with fibrin = > **bright pink red** amorphous appearance => **fibrinoid (fibrin like)**

Loss of normal structure & replacement by a homogenous, bright pink necrotic material that resembles fibrin microscopically



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► Reading Assignment

What is Necroptosis and how is different from apoptosis and necrosis?

